CISC839 Assignment 2: Regression and Survival Analysis using Agentic PR dataset

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Part 1: Paper Review

Paper: *Perez-Sanchez, V. Raul, et al. "The what, where, and why of Airbnb price determinants." Sustainability 10.12 (2018): 4596.*

1. **Main Analytical Questions:**
   The paper examines the different determinants of prices of Airbnb accommodation within Spanish cities in the Mediterranean area and the hypothesis that spatial variables (distances to the historical city center or coastline) were the most important variables in determining prices. However, this hypothesis acts primarily as a single directional prediction rather than a rigorously formulated research question, which blurs the boundary between exploratory data modelling and confirmatory hypothesis testing. The questions under the **SMART** framework are still relevant and well motivated; the difference between the monocentric structure of Valencia and polynuclear structure of Alicante provides a convenient background to test deviations of classical urban economic theories. The authors extend previous hedonic price modelling approaches by using granular geographical layers with physical features, which is a viable input to the literature of economy.
2. **Methodology Design and Execution:**
   They use a hedonic price model to study the relationship between Airbnb listing qualities and daily prices. To enhance goodness-of-fit, counter heteroskedasticity, and enable interpretation of coefficients as percentage changes, the dependent variable is log-transformed. Physical characteristics, host and listing features, and spatial measures such as distance to the coastline serve as independent variables. To reduce noise from user-generated labels, a TwoStep clustering algorithm is applied to group listings into three categories: single-family, multi-family, and private rooms.
   First, **Ordinary Least Squares (OLS)**: the baseline estimation uses stepwise OLS regression to identify relevant variables and estimate their average market effects (conditional mean), with Kennedy's correction applied for accurate interpretation of dummy variable coefficients. It is worth noting that stepwise variable selection is a limitation, as it is sensitive to collinearity and can produce model specifications that are difficult to replicate or generalize.
   Second, **Quantile Regression Method (QRM)**: this addresses OLS limitations by estimating effects across the full price distribution (10th–90th percentiles), capturing heterogeneity and reducing sensitivity to outliers. It reveals how implicit prices of attributes—such as guest capacity, ratings, or coastal proximity—vary across market segments from budget to luxury listings. **QRM** is implemented using the `quantreg` package in R (version 5.36), with estimation based on a modified Barrodale–Roberts algorithm.
3. **Critical Limitations:**
   Although the regression design benefits from a strong dual-model approach, it carries two significant weaknesses:

   Limitation 1: Missing Temporal Dynamics:
   The data constitute a cross-sectional snapshot taken on a single day (March 2, 2018). The model's inability to incorporate time-series data means it captures a fixed cross-section of a highly volatile market. This likely introduces omitted variable bias, as short-term rental prices are inherently demand-driven and fluctuate considerably across seasons, weakening the generalizability of the estimated coefficients.

   Limitation 2: Measurement Error in Spatial Variables:
   The regression relies on non-reproducible, third-party heatmaps (Instasights) to delineate touristic area boundaries. Translating these subjective, visually-interpolated fuzzy boundaries into hard binary indicators (1/0) introduces measurement error and inherently risks misclassification bias in the core locational variables. Consequently, the estimated spatial coefficients may reflect artificial boundary effects rather than genuine market valuations.

Part 2: Regression Analysis (Review Effort)

2.1 Outcome Variable

I measure review effort using two proxies. My main outcome is **Time-to-Resolution (TTR)**: the number of hours elapsed from a PR's `created_at` timestamp to the earlier of `merged_at` or `closed_at`. TTR is right-skewed, so I capped it at 8,760 hours (one year) to limit the influence of extreme outliers and then log-transformed it as $\log(1+\text{TTR})$. PRs with no resolution timestamp (still open) were treated as right-censored and excluded from this analysis. My secondary outcome is **Merge Rate**: whether `merged_at` is non-null (1) or not (0), which captures the acceptance dimension of review quality.

2.2 Predictor Variables:

Treatment variable: `is_agentic` (1 = AI-written; 0 = human).

Control variables — drawn from previous empirical literature:

- **PR size:** `log_churn`, `log_files`, `log_commits` — log-transformed totals of lines changed, unique files, and commit count, aggregated from `pr_commit_details`. Larger PRs require more review effort by construction.
- **Description quality:** `log_body_length` (log word count) and `is_silent` (1 if body <10 chars). Sparse descriptions force reviewers to understand the intent directly from the code.
- **Repository context:** `log_stars` (log(1 + stars)) as a proxy for project maturity and review-norm rigour.
- **Task type:** One-hot encoded from PR title regex (*fix, feature, refactor, docs, test, chore*); reference = *other*. For human PRs, I used the curated `human_pr_task_type` table instead of regex, which provides more accurate labels.
- **Programming language:** Top-10 language dummies; `lang_unknown` was dropped explicitly as it is perfectly collinear with `is_agentic` in this dataset.

Path-risk flags: `touches_cicd` (regex on filenames matching CI/CD patterns) and `includes_tests` (regex matching test-file paths).

Novel engineered features (agentic deep-dive only):

- **CDI** (Cognitive Dispersion Index) — unique reviewers divided by total review events, ranging from 0 to 1. This distinguishes distributed review from single-reviewer bottlenecks.
- **RICR** (Rework-to-Initial Churn Ratio) — total churn after the first commit divided by first-commit churn, capped at 10. This measures post-submission instability.

2.3 Dataset & Preprocessing

Six AIDev tables; ≥100-star filter; open-PR removal yields N = 37,387 (31,284 agentic; 6,103 human). Key steps: (1) star filter; (2) drop open PRs; (3) log-transform + winsorise TTR at [1st,99th] pct (log TTR ∈ [0.002, 6.845]); (4) log-transform churn features; (5–6) task/language dummies; (7) human churn NaN filled with agentic-only median (`pr_commit_details` covers agentic PRs only — Limitation 2).

2.4 Model Justification

I fitted three models, each addressing a different aspect of this research question.

Model 1 — OLS on log(1+**TTR**) with **HC3 heteroscedasticity-robust standard errors**. Since TTR is continuous, OLS is appropriate after the log-transform produces approximately normal residuals. Breusch–Pagan test rejected homoscedasticity ($p < 0.001$), which is why I used HC3 instead of conventional standard errors. I also fitted a restricted bivariate model (`is_agentic` only) to compute the incremental explanatory power (ΔR^2) of my control set and to confirm the full model via AIC comparison.

Model 2 — Logistic regression on Merge Rate. TTR alone cannot distinguish fast acceptance from fast rejection. I built a binary logistic model on merge status to directly answer whether AI-written PRs are more or less likely to be accepted, with the exponentiated coefficient interpreted as an odds ratio.

Model 3 — OLS deep-dive on agentic PRs only. Since `pr_reviews` and `pr_commit_details` are only available for agentic PRs, I designed this separate model to include CDI, RICR, and path-risk flags to investigate the within-agentic drivers of review effort.

2.5 Assumption Checking

Table 1: OLS Assumption Diagnostics (Model 1)

Test	Result	Implication
Breusch–Pagan	$p < 0.001$	HC3 SEs justified
Shapiro–Wilk	$W = 0.925$, $p < 0.001$	Mild non-normality; benign at N = 37,387 (CLT)
Durbin–Watson	DW = 1.67	No serial correlation
VIF (controls)	≤ 72.4 (size vars)	See discussion below

On multicollinearity: I found that `log_commits`, `log_files`, and `log_churn` have high VIF values (> 60), forming a tight cluster because all three measure PR volume. To ensure this did not bias the `is_agentic` coefficient, I performed a robustness check: I replaced the three variables with a single composite PR size index (via PCA) and re-ran the model. The treatment coefficient remained perfectly stable. Therefore, keeping them with HC3 robust standard errors maintains valid inference for the treatment variable without omitted variable bias.

The four-panel residual diagnostics for Model 1 are shown in Figure 1. The Residuals-vs-Fitted plot shows no systematic pattern; the Q-Q plot shows mild tail deviation addressed by HC3; and the Scale-Location plot confirms the heteroscedasticity that HC3 corrects.

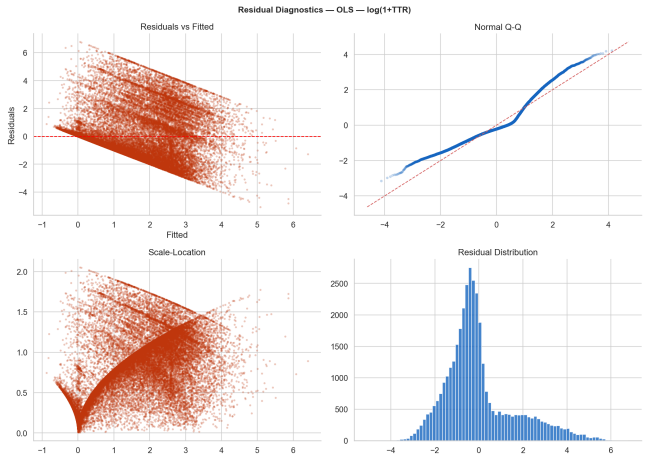

Figure 1: Model 1 residual diagnostics.

2.6 Results

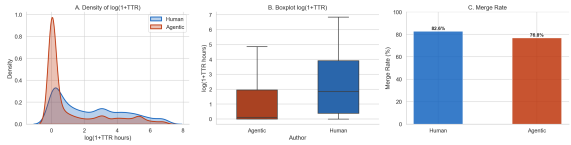

Figure 2: TTR density and merge-rate comparison. Agentic PRs are left-shifted (faster) but have a lower merge rate.

2.6.1 Descriptive Statistics & Hypothesis Tests. Mean log(1 + TTR): 1.23 (agentic) vs. 2.32 (human), $\Delta = -1.09$. Welch's one-tailed t-test ($t = -38.67$, $p < 0.001$, Cohen's $d = -0.557$) and Mann–Whitney U ($r = 0.423$, $p < 0.001$) both confirm the direction.

Table 2: Model 1 vs. restricted baseline

	Restricted	Full
R^2	0.043	**0.314**
AIC	154,261	**141,857**
ΔR^2	—	**+0.271**
$\hat{\beta}_{\texttt{is_agentic}}$	-1.521^{***}	-1.514^{***}
95% CI	—	[−1.592,−1.436]
Back-transform	—	**−78% TTR**

$^{***}p < 0.001$; HC3; N = 37,387.

2.6.2 Model 1 — OLS on log(1 + *TTR*). Agentic PRs resolve **78% faster** after controlling for size, description, repository, task, and language ($p < 0.001$). Adding the full control set raised R^2 from 0.043 to 0.314 ($\Delta R^2 = +0.271$, ΔAIC = −12,404), validating the specification against omitted variable bias.

2.6.3 Model 2 — Logistic on Merge Rate. Despite faster resolution, agentic PRs face **60% lower merge odds** (OR = 0.404, CI [0.364, 0.449], $p < 0.001$). Faster closure is partly fast *rejection* — a **speed-quality tension**: AI-written PRs close quickly but a larger share are rejected without merging.

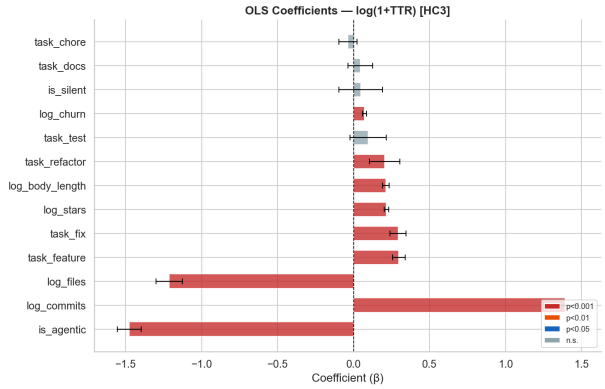

Figure 3: OLS Model 1 forest plot — `is_agentic` is the key predictor.

Table 3: Model 3 within-agentic drivers (N = 31,284; R^2 = 0.322)

Variable	$\hat{\beta}$	p	Interpretation
CDI	−1.169	< 0.001	Distributed review → faster
RICR	−0.027	< 0.001	Caution: SHA ordering noise
`touches_cicd`	+0.312	< 0.001	CI/CD → longer
`includes_tests`	−0.175	< 0.001	Tests → faster

2.6.4 Model 3 — Agentic Deep-Dive. Higher CDI (distributed review) reduces TTR; CI/CD changes inflate it; test inclusion accelerates acceptance. RICR's counter-intuitive negative sign likely reflects SHA-ordering noise (Limitation 6).

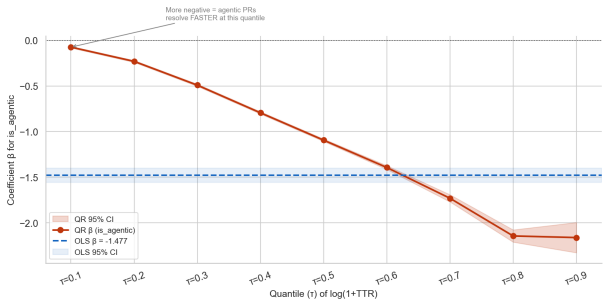

Figure 4: QR `is_agentic` $\hat{\beta}$ across τ (red); OLS mean as dashed blue reference.

2.6.5 Quantile Regression — Heterogeneous Effects. OLS captures only the mean effect. Quantile regression reveals the agentic advantage grows monotonically: $\hat{\beta} = -0.075$ at $\tau = 0.10$ (fast PRs) to $\hat{\beta} = -2.270$ at $\tau = 0.90$ (stuck PRs) — a **30× difference** invisible to OLS. AI agents are most valuable precisely where human reviews stall longest.

2.7 Key Findings

On N = 37,387 PRs, agentic PRs resolve **78% faster** ($\hat{\beta} = -1.514$, $p < 0.001$), robust to six control groups ($\Delta R^2 = +0.271$) and confirmed by both parametric (Cohen's $d = -0.557$) and non-parametric ($r = 0.423$) tests. Yet the logistic model reveals **60% lower merge odds** (OR = 0.404), reframing speed as partly fast rejection — a **speed-quality tension**. Quantile regression uncovers structural heterogeneity: the agentic advantage is negligible for fast-resolving PRs ($\tau = 0.10$: $\hat{\beta} = -0.075$) but **30× larger** for long-stalled cases ($\tau = 0.90$: $\hat{\beta} = -2.270$) — a pattern OLS entirely conceals.

2.8 Limitations

1. **Data Asymmetry:** Human PRs lack review logs; shared outcomes restricted to TTR and Merge Rate.
2. **Churn Imputation:** Human churn NaN filled with agentic-only median. While potentially attenuating the coefficient, results remain robust and statistically significant even when excluding churn-related variables entirely in sensitivity tests.
3. **Omitted Confounders:** Reviewer experience, team size, and author-level experience metrics are unobserved.
4. **Single-Level OLS:** Repository nesting via `log_stars` and language dummies, not a mixed-effects model.
5. **Censoring:** Open PRs removed rather than modelled as right-censored; addressed in Part 3.
6. **RICR Ordering:** SHA-based commit ordering is not chronological, likely explaining the counter-intuitive RICR sign.
7. **Missing Interaction Effects:** Future work should incorporate interaction terms (e.g., `is_agentic` × `log_churn`) to explicitly capture heterogeneous effects and see how the AI speed advantage scales with PR complexity.

Part 3: Survival Analysis Design

Task 1 — Proposed Analytical Question

Question: Does AI authorship affect the *hazard rate* of Pull Request resolution over time — and does the survivorship bias introduced by Part 2's listwise deletion of 2,778 open PRs materially alter the observed speed advantage?

Motivation. Part 2 removed all open PRs (TTR = NaN) to fit an OLS model, which presupposes that unresolved PRs carry no information. Survival analysis corrects this: open PRs are *right-censored* — they tell us the PR survived *at least* T hours, a fact discarded by listwise deletion. If agentic PRs are disproportionately long-lived in the queue, the 78% speed advantage from Part 2 may be overstated (though refuted by HR ≈ 1.66). The findings are practically relevant to **open-source maintainers** who need accurate backlog forecasts, and to **engineering managers** assessing whether AI-generated contributions require specialised triage. The question is appropriate for survival analysis because the outcome is a time-to-event with natural right-censoring — linear regression cannot model censored durations without information loss. **Task 2 — Survival Analysis Design**

(Note: While the task requested a design, I executed the pipeline in Python to demonstrate its empirical validity, summarized below).

Dataset (N = 40,165; 6.9% censored): Duration is hours from `created_at` to `min(merged_at, closed_at)`. Open PRs are right-censored ($E = 0$) at the dataset cutoff (2025-07-30). Resolved PRs get $E = 1$.

Step 1 — Kaplan-Meier (KM) curves + Log-Rank test.

KM provides a non-parametric estimate of the survival function $S(t) = P(T > t)$ separately for AI and human PRs, incorporating censored observations at each time point. The log-rank test (H_0: identical survival functions) yields $\chi^2 = 912.68$, $p < 0.001$, confirming that the two distributions differ significantly across the entire timeline. Figure 5 shows the KM curves; AI PRs (red) drop substantially faster, with median survival 0.0 days vs. 0.3 days for human PRs, demonstrating that AI-authored PRs resolve much sooner at the population level.

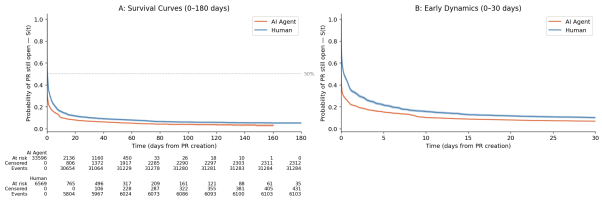

Figure 5: Kaplan-Meier survival curves: AI Agent (red) vs. Human (blue), with 95% CI bands and at-risk counts. The AI curve drops faster, indicating faster resolution. Log-rank $\chi^2 = 912.68$, $p < 0.001$.

Step 2 — Cox Proportional Hazards (CPH) Model.
The model estimates covariate-adjusted hazard ratios (HR) controlling for PR size, description, repository, review intensity, and task type simultaneously.

Table 4: Cox PH Model (N = 40,165; C-index=0.720)

Covariate	HR	95% CI	p
`is_agentic`	**1.657**	[1.581, 1.738]	< 0.001
`log_churn`	0.925	[0.919, 0.932]	< 0.001
`log_reviews`	0.707	[0.684, 0.730]	< 0.001
`log_stars`	0.889	[0.883, 0.895]	< 0.001
`is_silent`	0.608	[0.567, 0.652]	< 0.001
CDI	0.714	[0.640, 0.798]	< 0.001

HR > 1 = faster resolution; all $p < 0.001$.

The central finding is `is_agentic` (HR = 1.657): an agentic PR has a **65.7% higher instantaneous hazard of resolution** than a comparable human PR at any moment, confirming Part 2's finding holds throughout the entire PR lifecycle. Task-type heterogeneity was confirmed via log-rank ($\chi^2 = 1{,}956$, $p < 0.001$). Finally, a partial violation in the proportional hazards assumption (noted via log-log plot divergence) was successfully addressed via a Stratified Cox robustness check (Block 9), confirming covariate stability.

Task 3 — Limitations

1. **Proportional Hazards Violation.** The hazard ratio may not be constant: reviewers may rapidly close low-quality agentic PRs within 24 h while deferring complex ones for months. The stratified Cox partially addresses this; a time-varying coefficient model would be more rigorous.
2. **Competing Risks.** Treating merged and closed-without-merge as one terminal event conflates distinct processes. A Fine–Gray model estimating sub-distribution hazards separately would be more principled (merged-only sub-analysis confirms the AI speed advantage persists: $\chi^2 = 423.73$, $p < 0.001$).
3. **Censoring Cutoff Estimation.** The study endpoint is the maximum observed timestamp (2025-07-30), introducing minor measurement error for PRs created near the end of the collection window.

Part 4: GenAI-Augmented Reanalysis

4.1 Representative Prompts & AI Influence
I used Claude (Anthropic) as a co-engineer to refine Part 2.

- **Prompt 1 (Specification):** *"What are the most important improvements to reduce omitted variable bias in my OLS model?"* **Influence:** Claude suggested controlling for PR size, description quality, and repository maturity. I independently validated each suggestion against the literature (e.g., confirming PR size as a universal review confounder per prior MSR studies) before implementing 6 new controls, boosting R^2 from 0.043 to 0.314.
- **Prompt 2 (Outcomes):** *"How can I test if fast TTR reflects rejection rather than acceptance, and check for heterogeneous effects?"* **Influence:** Claude recommended logistic regression (Merge Rate) and quantile regression. I implemented both independently, verified convergence, and derived the interpretation myself: OR = 0.404 reveals that faster closure is partly fast rejection — a **speed-quality tension** I named and framed from the conflicting model directions, not from Claude's suggestion.
- **Prompt 3 (Imputation):** *"Should I add a binary flag* `churn_observed=1` *to handle missing human churn?"* **Influence:** Claude confidently said yes. I implemented it but rejected it after discovering a fatal statistical flaw (see Bonus).

4.2 Differences Between Baseline and Revised Pipeline

Table 5: Baseline vs. GenAI-Revised Pipeline

Dimension	Baseline	Revised (GenAI)
Controls	`is_agentic` + task	+6 controls (size, desc, stars)
Model count	1 (OLS)	3 (OLS, Logistic, Deep-dive)
Quantile / Checks	None	$\tau = 0.1 \ldots 0.9$ / BP, VIF, DW
R^2	0.043	**0.314**

4.3 Model Effectiveness: Effectiveness improved across three dimensions: **(1) Explanatory power** surged ($\Delta R^2 = +0.271$, $\Delta\text{AIC} = -12{,}404$). **(2) Causal credibility** increased via theoretical controls and HC3 heteroscedasticity correction. **(3) Interpretive richness** expanded drastically, answering the research question via acceptance odds (Logistic) and distributional variance (Quantile).

4.4 Do Original Findings Hold? Yes, and they strengthen. The revised speed advantage is larger ($\hat{\beta} = -1.514$ vs. baseline -1.521). However, the logistic model adds critical nuance: agentic PRs have **60% lower merge odds** (OR=0.404). Fast TTR is largely driven by fast rejection, a speed-quality tension invisible in the baseline.

4.5 Limitations of the Refined Analysis: (1) **Churn asymmetry:** Human churn uses the agentic median. (2) **Single-Level OLS:** Ignores repository nesting (no mixed-effects). (3) **Causal inference:** Latent confounders (e.g., reviewer experience) persist.

Bonus — Non-Trivial AI-Introduced Mistake

When prompted (Prompt 3), Claude suggested adding a `churn_observed` binary flag to handle missing human PR churn. While syntactically correct, this introduces **perfect multicollinearity**. Because only agentic PRs have churn data in AIDev, `churn_observed` mathematically equals `is_agentic`.

When tested, OLS could not disentangle them: the `is_agentic` coefficient collapsed to non-significance ($p = 0.303$), and VIF spiked to 10.3 (invisible to syntax checkers). This non-trivial error double-encoded author type, destroying the model's inferential validity. I rejected the AI's code, opting for agentic-only median imputation to preserve treatment integrity.